



every step emits one record

One telemetry record per step, written before the step can count

step scalars

reward

loss

entropy

KL to ref. policy

grad norm

group diagnostics

ZVF

gradient utilisation
 $1 - \text{ZVF}$

all-correct
fraction

all-wrong
fraction

behaviour

mean completion length

clip fraction

learning rate (and schedule)

provenance

step

seed

config
hash

commit
SHA

framework
id

grader
id

Raw per-group reward tensor $\mathbf{r}$, of shape $|B_t| \times G$, is retained, not discarded. The scalar ZVF above is only a summary: the tensor behind it is kept, so the all-correct / all-wrong split can be recomputed and re-audited after the run has ended.

written to

Local JSONL

one record per step, append-only; analysable offline after the run

written to

Weights & Biases

the same record mirrored for live monitoring and external review

written to

Checkpoint $\rightarrow$ Hugging Face
published with its revision URL, paired with the records behind it

Consumers of the record

Recomputation.

`p2_collapse_analysis.py` reruns ZVF and the all-correct / all-wrong split from the retained tensors. The published P2 values (0.130, 0.190, 0.155; pooled 0.1583) come from this path, not from the live training loop.

Normalisation. One ZVF definition. A normaliser maps the reward and mask fields of four frameworks (TRL, Tinker, OpenRLHF, verL) onto it, so a ZVF reported for one of them means the same as for another.

Receipt and ledger. Freeze-before-paid-work: the record shows what a run actually did, and it is what the seven-state lane taxonomy and the audit suite are scored against.

Fail closed. A step that cannot write its record halts the run. Telemetry is a precondition of progress, not a post-hoc summary written from memory — which is also why ZVF is defined on the reward *spread* rather than on a hand-annotated failure mode.